EcoDrive

Project Team

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Session 2021-2025

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June, 2025

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Chapter 1

Introduction

EcoDrive is an innovative waste management system designed to address the inefficiencies of traditional waste collection methods. By utilizing a network of smart bins and an autonomous vehicle, EcoDrive optimizes waste collection routes and schedules, leading to significant improvements in efficiency, cost reduction, and environmental impact. The system is designed to overcome the limitations of fixed scheduling in existing waste collection services, where waste is often collected regardless of whether the bins are full or not. This leads to wasted resources and an inefficient use of manpower. EcoDrive aims to solve this problem by enabling on-demand waste collection, triggered by the smart bins when they reach their full capacity.

1.1 Research on Existing Products

Previous research in waste management systems focuses on improving the efficiency of waste collection in urban areas through real-time monitoring and data collection. A notable system involves the use of smart bins equipped with sensors that measure waste levels, weight, temperature, and humidity. The data collected enables municipal authorities to optimize waste collection schedules and routes, reducing fuel consumption and enhancing overall efficiency. The smart bin system utilizes ultrasonic sensors to measure waste levels and load cells to monitor weight, ensuring accurate detection regardless of waste type. Additionally, a temperature and humidity sensor provides environmental data, enhancing user convenience. Through GPS tracking, the system pinpoints bin locations, allowing authorities to efficiently plan waste collection. This system also promotes public engagement by enabling citizens to monitor nearby bin statuses in real time, preventing waste overflow and contributing to cleaner urban environments. This project includes contributions from Syed Ali Hassan, who is affiliated with the National University of Sciences and Technology (NUST), Islamabad, Pakistan, as part of an international collaboration aimed at improving waste management solutions. Furthermore, other research efforts conducted by the School of Electronics Engineering, Kalinga Institute of Industrial Technology (Deemed to be University), Bhubaneswar, India, emphasize the integration of Radio Frequency Identification (RFID), GPRS, GPS, and Geographic Information System (GIS) technologies for waste collection optimization. These technologies allow for automatic bin identification, real-time tracking, and location mapping, further streamlining the waste collection process and ensuring more efficient resource allocation. [1]

1.2 Problem Statement

The current waste management practices suffer from inefficiencies due to fixed scheduling, leading to unnecessary trips, wasted resources, and potential environmental hazards. Overfilled bins can attract pests, contribute to the spread of diseases, and cause blockages in drainage systems.

1.2.1 Business Opportunity

EcoDrive presents a business opportunity to revolutionize the waste management industry by offering a more efficient, cost-effective, and environmentally friendly solution. The system has the potential to reduce fuel consumption, labor costs, and the overall environmental impact of waste collection. Inefficient waste management practices are a major global issue, costing the world economy hundreds of billions of dollars annually. In 2020, the World Bank estimated that inefficient waste management cost up to $410 billion globally, due to factors such as increased labor, transportation, and landfill fees. In the U.S. alone, cities and municipalities often face significant costs due to inefficient waste management. For example, municipalities may spend $50 to $100 per ton on waste disposal, with poor management practices leading to an increase of 20-50% in these costs. This results in billions of dollars in additional expenses each year. By optimizing waste collection routes and enabling on-demand services, EcoDrive has the potential to drastically reduce these municipal costs. Moreover, inefficient waste management contributes significantly to environmental damage. In the U.S., environmental costs related to health impacts and pollution caused by poor waste practices amount to $375 billion annually. EcoDrive can play a crucial role in reducing these environmental costs by preventing overfilled bins, reducing pollution, and improving public health. Another critical issue is the loss of valuable resources due to inadequate recycling and sorting practices. Globally, around $200 billion worth of recyclable materials are lost to landfills each year. By incorporating smart bin technology and optimizing waste management systems, EcoDrive can help recover these lost resources and promote sustainability.

1.2.2 Objectives

- To develop a fully functional autonomous waste collection system.
- To optimize waste collection routes and schedules.
- To reduce the environmental impact of waste management.
- To improve the efficiency and cost-effectiveness of waste collection.

1.3 Scope

The EcoDrive project focuses on the development and testing of a proof-of-concept autonomous waste collection system within a predefined environment. This environment will feature strategically placed smart bins that monitor their fill levels using various sensors and signal the autonomous vehicle when they reach capacity. The vehicle, a scaled-down autonomous toy truck, will leverage self-driving technology to navigate the environment autonomously, collecting trash from the smart bins along the most optimal route. By showcasing how self-driving technology can be applied to waste collection, EcoDrive highlights the potential for transforming traditional waste management practices, making them more efficient, cost-effective, and sustainable.

1.4 Constraints

- The project is limited by the availability of resources and time constraints.
- The system's performance may be affected by external factors such as weather conditions and traffic congestion.
- The initial deployment of the system will be limited to a controlled environment.

1.5 Modules

1.5.1 Module 1: Smart Bin Module

This module handles the functionality of the smart bins, including sensor data collection and communication with the central server.

1.5.1.1 Level Sensors

- Detect the fill level of the bins.
- Trigger a signal when the bin is full.
- Communicate with the central server when the bin reaches capacity.

1.5.1.2 Lid Automation

- Automatically opens the bin lid when a person approaches to throw trash.
- Ensures smooth operation through a motion sensor or proximity sensor.

1.5.1.3 Communication Subsystem

- Transmits bin status to the central server in real-time using a secure communication protocol.
- Monitors bin health and functionality for maintenance notifications.

1.5.2 Module 2: Central Server and Communication Module

This module manages all the communication between the smart bins, autonomous trucks, and the dashboard. It acts as the system's brain.

1.5.2.1 Signal Processing

- Receives signals from smart bins (capacity reached) and relays them to the autonomous truck for collection.
- Manages and coordinates real-time communication between smart bins and trucks.

1.5.2.2 Data Aggregation

- Collects data from all smart bins for monitoring purposes.
- Prepares the data for dashboard display, providing real-time fill level updates and navigation status.

1.5.2.3 Security Subsystem

Implements encrypted communication channels between bins, trucks, and the server. Role-based access control to restrict unauthorized access.

1.5.3 Module 3: Autonomous Vehicle (Toy Truck) Module

This module focuses on the navigation, perception, and decision-making aspects of the self-driving truck.

1.5.3.1 Perception Subsystem

- Cameras and LiDAR: Capture visual and distance data from the environment to detect obstacles and aid in navigation.
- Sensor fusion combines data from multiple sensors for enhanced perception accuracy.

1.5.3.2 Navigation Subsystem

- Uses algorithms like reinforcement learning (RL) for real-time route optimization.
- Processes sensor data and computes optimal routes to the bins and the waste facility.
- Obstacle Detection: Identify and avoid obstacles using LiDAR and camera inputs.

1.5.3.3 Decision-Making Subsystem

- End-to-end decision-making process powered by RL to autonomously navigate and collect trash.
- Uses reinforcement learning to learn and adapt to the environment over time.

1.5.3.4 Communication with Central Server

- Receives dispatch signals from the central server.
- Sends navigation status updates and completed task signals to the server.

1.5.3.5 Security Subsystem

- The Jetson Nano in the truck is protected by a firewall and ensures secure communication with the central server.

1.5.4 Module 4: Dashboard and User Interface Module

This module provides a web-based interface for monitoring and controlling the system.

1.5.4.1 Real-Time Monitoring

- Displays the fill level of all smart bins and the navigation status of the autonomous truck.
- Provides visual indicators for truck position, bin status, and waste facility status.

1.5.4.2 User Control Interface

- Allows users to control and monitor the system, including truck dispatch, bin health status, and more.
- User-friendly interface with accessible design (screen reader compatibility, high contrast mode).

1.5.4.3 Data Visualization

- Presents real-time analytics such as bin fill levels, collection times, and overall system performance.
- Offers historic data visualization for system performance analysis.

1.5.4.4 Security

- Role-based access control for different users (e.g., maintenance team, system administrators).

- Logs all access attempts for auditing.

1.5.5 Module 5: Safety and Compliance Module

This module ensures the system adheres to safety standards and functional safety protocols.

1.5.5.1 ISO 26262 Compliance

Ensure that all self-driving components comply with ISO 26262, focusing on the functional safety of road vehicles.

1.5.5.2 Emergency Handling

Implements fail-safe mechanisms to stop the truck in the event of system failures or emergencies. Regular diagnostics and maintenance alerts for hardware and software systems.

1.5.6 Module 6: Maintenance and Scalability Module

This module ensures the long-term reliability and scalability of the system.

1.5.6.1 System Maintenance

- Supports easy updates to hardware and software with minimal downtime.
- Monitors system health and alerts the team when maintenance is required.

1.5.6.2 Scalability Subsystem

- Allows the addition of more smart bins and trucks without overloading the central server.
- Optimizes data storage and processing for future system expansions.

1.6 User Classes and Characteristics

Stakeholder	Goals	Problems
Municipal Corporations	Efficient waste management, cost reduction, improved public health	Inefficient waste collection, overflowing bins, environmental concerns
Development Authorities	Sustainable development, improved infrastructure	Waste management challenges, environmental impact
Educational Institutions	Clean and healthy campus environment, sustainability initiatives	Waste disposal issues, environmental awareness
Waste Management Companies	Improved efficiency, cost reduction, competitive advantage	Rising operational costs, labor shortages, environmental regulations
Environmental Agencies	Environmental protection, reduced pollution	Waste-related pollution, greenhouse gas emissions
Local Communities	Clean and healthy living environment, improved quality of life	Overflowing bins, health hazards

Table 1.1: Key High-Level Goals and Problems of Stakeholders

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Event Response Table

Event ID	Event	Response
EV-001	Bin reaches capacity	Signal sent to the server with bin ID and capacity status.
EV-002	Signal received from bin	Process signal and assign truck to collect waste from the specified bin.
EV-003	Truck receives assignment	Truck begins navigation to the full bin based on optimal route calculation.
EV-004	Obstacle detected in truck's path	Autonomous Toy Truck recalculates route to avoid the obstacle.
EV-005	Truck arrives at full bin	Signal sent to the server that the truck has arrived; truck opens the bin lid, collects trash, and confirms collection.
EV-006	Bin lid opened automatically	Bin lid opens automatically when the user approaches to throw trash, and closes after a short period.
EV-007	Truck completes trash collection	Signal sent to the central server confirming completion of trash collection.
EV-008	Truck arrives at waste disposal site	Signal sent to the central server; truck disposes collected trash and resets for the next task.
EV-009	Truck requires maintenance	Signal sent to the central server if the truck detects hardware issues (e.g., battery low, malfunctioning sensor).
EV-010	Manual override requested	Central server or authorized user takes control of the truck in case of emergency or for testing.
EV-011	Dashboard request for bin or autonomous toy truck status	System retrieves and displays real-time status of bins and truck navigation on the dashboard.
EV-012	Autonomous Truck route recalculation required	System recalculates the optimal route for the truck based on road conditions, traffic, or obstacles detected by sensors.
EV-013	Bin sensor failure	Signal sent to the central server reporting sensor failure; maintenance team is notified for troubleshooting.
EV-014	Real-time system performance monitoring	Logs system status and sends performance metrics (truck speed, bin capacity, etc.) to the central server for analysis and reporting.

Table 2.1: Event-Response Table for EcoDrive Project

2.2 Functional Requirements

2.2.1 Module 1: Smart Bin Module

2.2.1.1 Bin Capacity Monitoring

The smart bin shall detect when it is approaching full capacity and trigger a notification when the bin reaches a pre-defined threshold (e.g., 90% full). The bin shall continuously monitor its capacity and update its status to the central system at regular intervals.

2.2.1.2 Lid Operation

The bin's lid shall automatically open when someone approaches to dispose of waste. The bin shall close the lid after a designated period of inactivity to prevent unauthorized access or environmental interference.

2.2.1.3 Communication

The bin shall transmit its capacity and operational status (e.g., sensor functioning, lid movement) to the central server in real-time. In case of communication failure, the bin shall store status information locally and transmit it when the connection is re-established.

2.2.2 Module 2: Central Server and Communication Module

2.2.2.1 Signal Processing

The central system shall process signals received from the smart bins and determine which bins require collection based on their capacity status. The central system shall track the status of all smart bins and update the status dashboard in real time.

2.2.2.2 Task Assignment

Upon receiving a full-bin signal, the central system shall assign the waste collection task to the autonomous vehicle. The system shall optimize task assignment based on truck location and bin priority to ensure efficient collection routes.

2.2.2.3 Monitoring and Logging

The system shall monitor communication with all smart bins and log their status for later analysis. The system shall log all data transfer and actions taken, including successful and failed communications.

2.2.3 Module 3: Autonomous Vehicle Module

2.2.3.1 Navigation

The vehicle shall autonomously navigate to designated bins using sensor inputs and predefined routes. The vehicle shall avoid obstacles and adjust its route dynamically based on the current environment and bin locations.

2.2.3.2 Waste Collection

The vehicle shall align with the smart bin and collect the waste efficiently after receiving a command from the central system. The vehicle shall signal to the central system once the waste collection process is completed.

2.2.3.3 Communication

The vehicle shall continuously communicate its location and task status to the central system during waste collection. In case of communication failure, the vehicle shall attempt to reconnect and transmit stored status information once the connection is restored.

2.2.4 Module 3: Dashboard/Monitoring Module

2.2.4.1 Status Monitoring

The dashboard shall display the real-time status of all smart bins, showing their fill level and operational state. The dashboard shall display the location and status of the autonomous vehicle during its waste collection task.

2.2.4.2 Alerts and Notifications

The system shall trigger alerts on the dashboard when bins are full or if there are any system malfunctions (e.g., sensor failures, vehicle issues). The dashboard shall provide notifications to maintenance staff if a bin or vehicle requires manual intervention.

2.2.4.3 Data Reporting

The system shall generate periodic reports on waste collection performance, vehicle operation, and bin usage trends. Historical data about bin capacity and collection routes shall be accessible via the dashboard for future analysis.

2.2.5 Module 5: Data and System Management Module

2.2.5.1 Data Storage

The system shall securely store data from the smart bins and the vehicle, including their operational status, capacity, and logs of all actions.

2.2.5.2 Access Control

The system shall implement role-based access control to restrict user actions based on their role (e.g., admin, user, maintenance).

2.2.5.3 System Updates

The system shall allow administrators to perform over-the-air updates to the smart bins, vehicle software, and dashboard without causing disruption to operations. These functional requirements define the core functionality of each module, ensuring that they work together to meet the overall system objectives of EcoDrive.

2.3 Non-Functional Requirements

2.3.1 Reliability

- RELI-1: The system shall have a Mean Time Between Failures (MTBF) of at least 200 hours.
- RELI-2: In the event of a software failure, the system shall log the failure details for analysis and provide a notification to the maintenance team within 5 minutes.
- RELI-3: The system shall implement automated error detection algorithms that will trigger corrective actions to minimize downtime.
- RELI-4: The system shall adhere to safety standards for self-driving vehicles, such as ISO 26262, which ensures functional safety of road vehicles.

2.3.2 Usability

- USAB-1: The web dashboard shall allow users to monitor bin status and truck navigation with a user-friendly interface that requires no more than two clicks to access essential functions.
- USAB-2: The system shall provide a tutorial for new users, ensuring that they can operate the interface effectively within 10 minutes of first use.
- USAB-3: The interface shall support accessibility features to accommodate users with disabilities, including screen reader compatibility and high-contrast mode.

2.3.3 Performance

- PERF-1: The system shall respond to bin capacity signals and relay information to the truck within 2 seconds.
- PERF-2: The autonomous vehicle shall process sensor data from cameras and LiDAR to make navigation decisions with a latency of no more than 100 milliseconds.
- PERF-3: The system shall successfully complete 90% of waste collection operations without human intervention within the first 30 minutes of deployment.

2.3.4 Security

- SEC-1: The system shall use encrypted communication for data transmitted between smart bins, the central server, and the autonomous truck to ensure data confidentiality.
- SEC-2: The Jetson Nano shall have its own firewall configured to protect the autonomous truck from external threats.
- SEC-3: An end-to-end decision-making approach shall be employed in the system, significantly reducing the chances of a security breach.

2.4 Domain Model

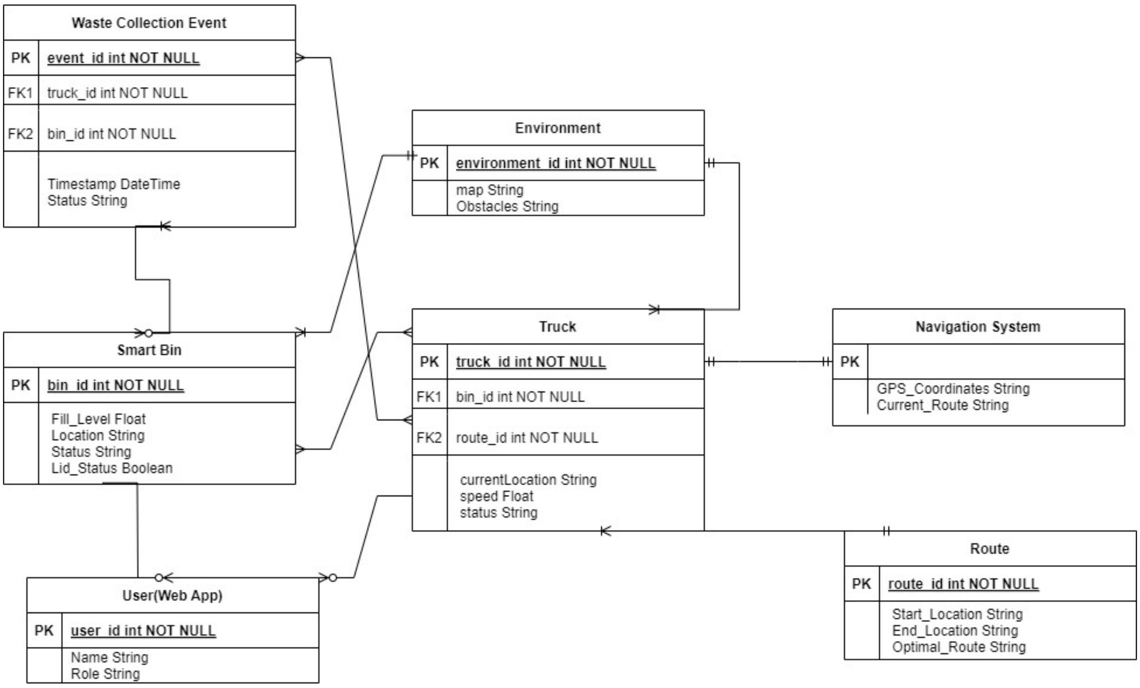

Figure 2.1: Domain Model

Chapter 3

System Overview

The EcoDrive project is an innovative autonomous waste collection system designed to improve urban waste management by integrating smart bins with an autonomous vehicle. The system aims to streamline the process of waste collection by using sensors to detect bin capacity and autonomous navigation to collect and dispose of waste. The smart bins, equipped with ultrasonic sensors, communicate with a central server when they are full. The server then dispatches a toy truck equipped with cameras and LiDAR to autonomously navigate to the bins and collect the waste.

This system operates in a controlled environment and includes various hardware and software components, such as the Jetson Nano for computation in the truck, ultrasonic sensors in bins, a central communication server, and a web dashboard for monitoring. EcoDrive also emphasizes scalability, reliability, and security, ensuring the solution can grow in the future while maintaining safety and operational standards.

3.1 Architectural Design

The system architecture is based on a modular design, allowing individual components to interact in a cohesive manner while maintaining independent functionality. EcoDrive is designed using the Client-Server Architecture style, where the smart bins, autonomous vehicle, and dashboard act as clients, and the central server coordinates the communication between these modules.

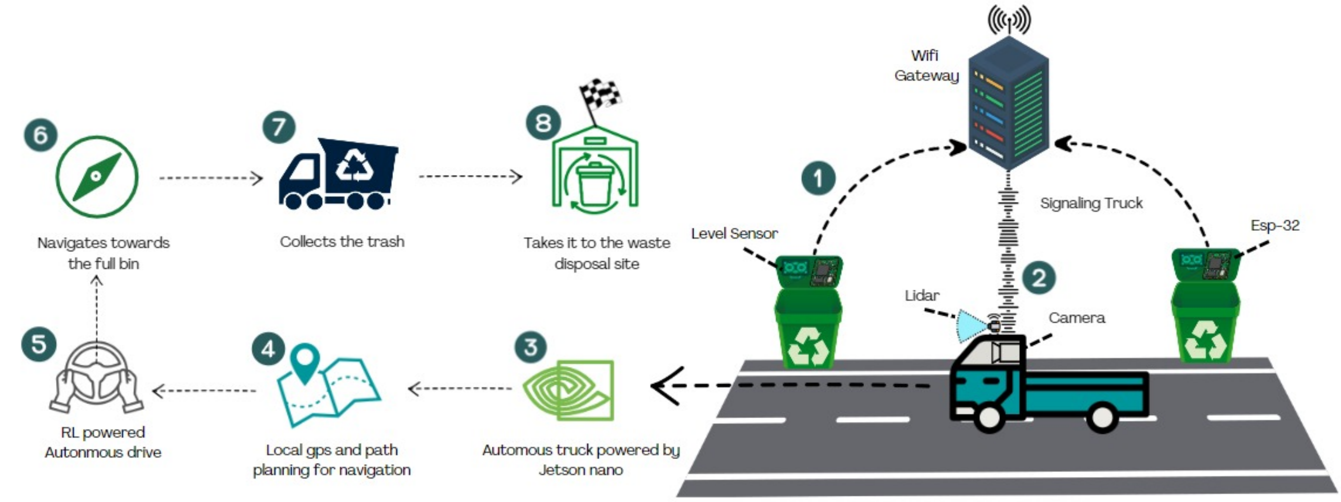

Figure 3.1: Architecture Diagram

3.2 Design Models

- Class Diagram

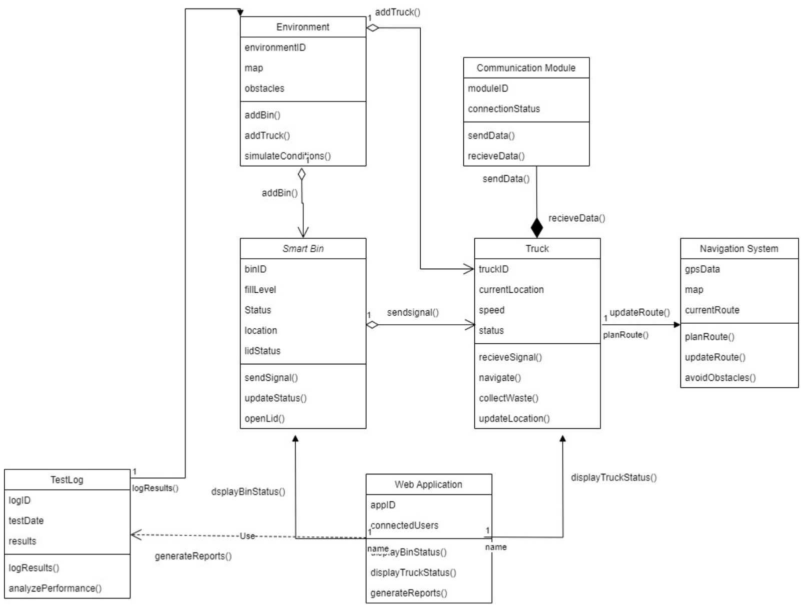

Figure 3.2: Class Diagram

- Activity Diagram

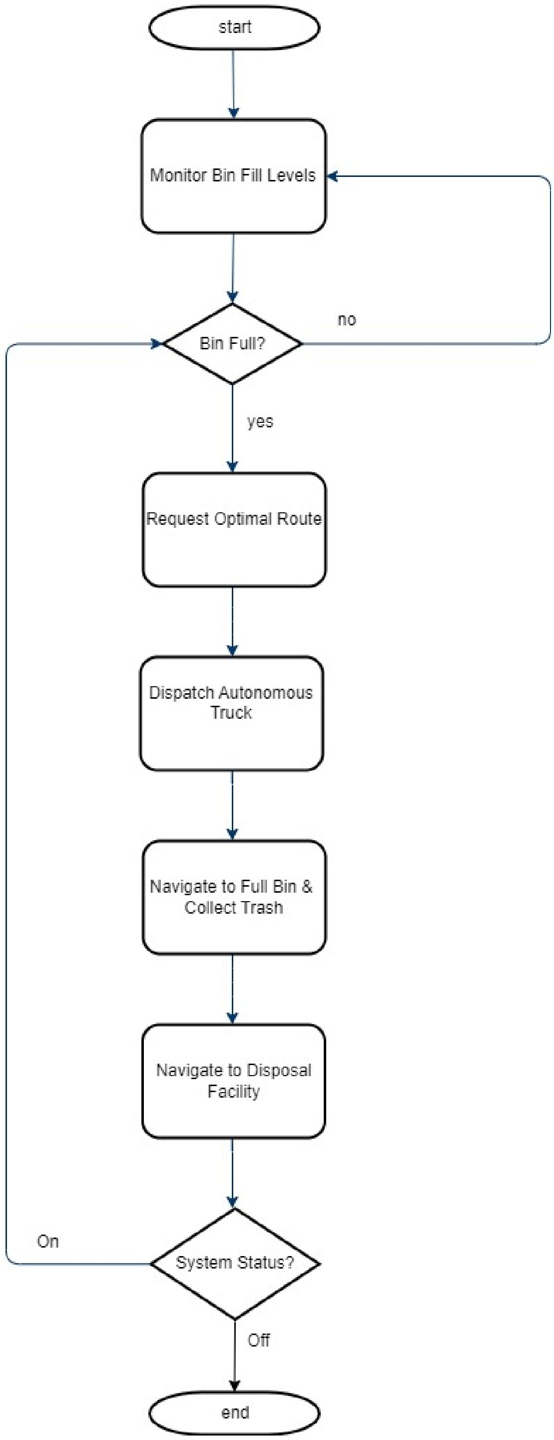

Figure 3.3: Activity Diagram

- Sequence Diagram

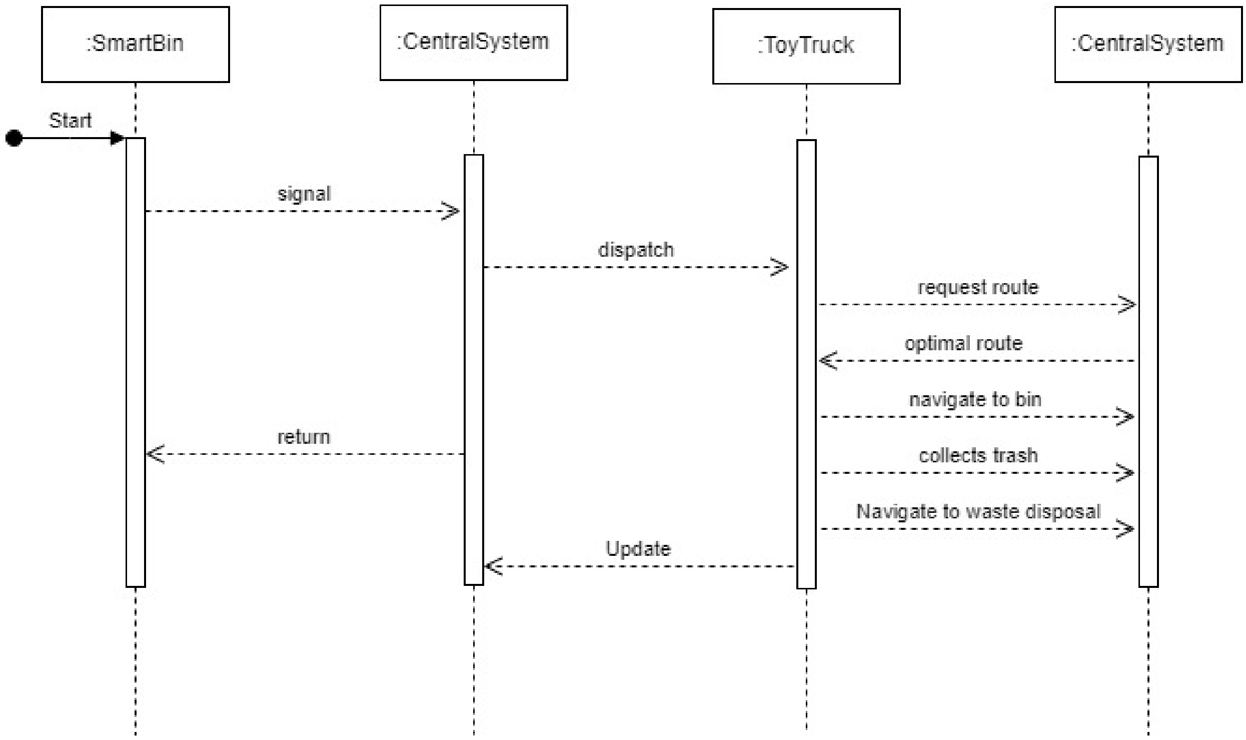

Figure 3.4: Sequence Diagram

- UI Screen

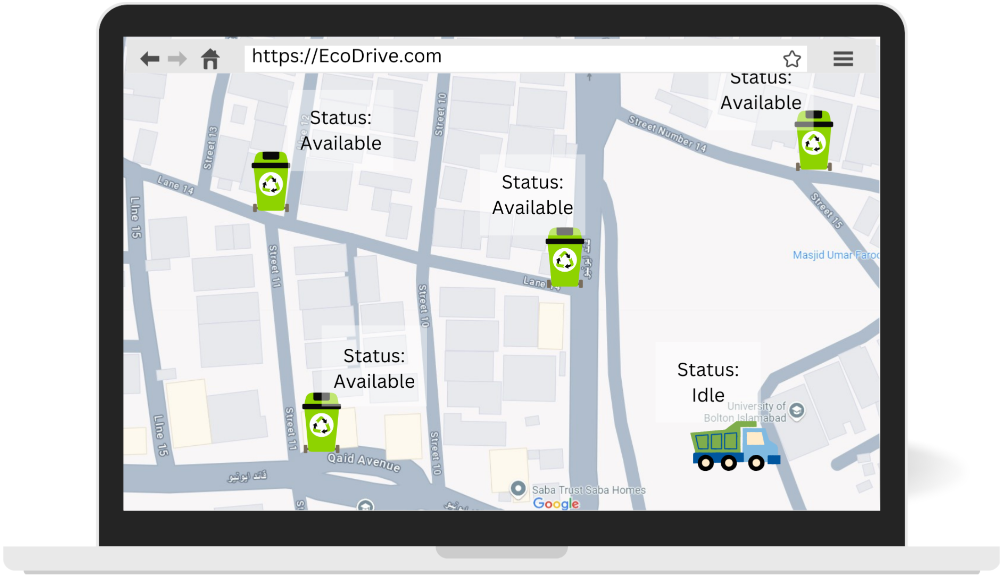

Figure 3.5: UI Screen-1

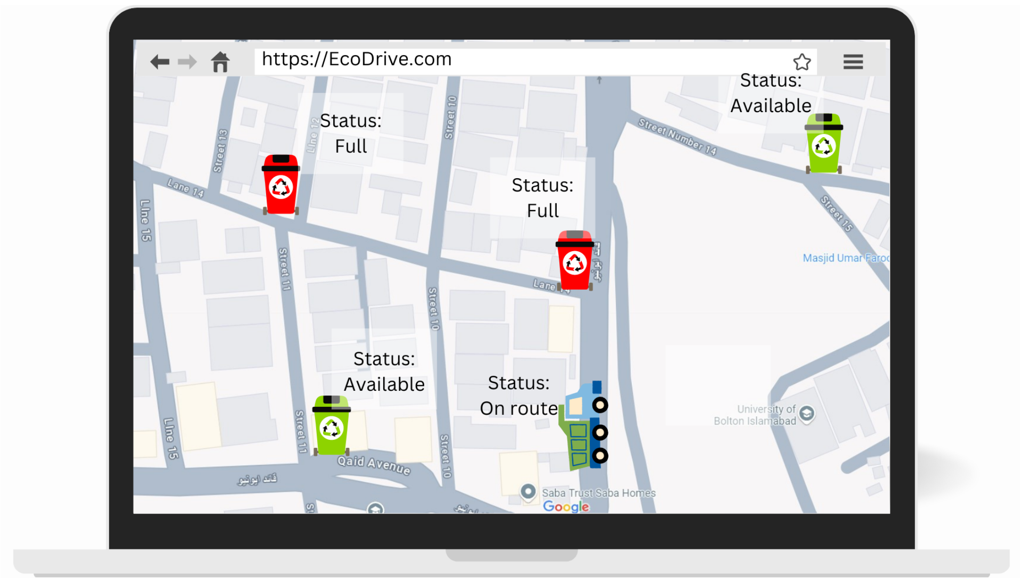

Figure 3.6: UI Screen-2

3.3 Data Design

In this project, the data generated and used by the system will be stored locally on the Jetson Nano device, which will handle both storage and computation. The major data entities in the system will include:

Bin Status Information: Each smart bin will communicate its fill level to the autonomous vehicle via the central server. This data will be temporarily stored on the Jetson Nano for processing and decision-making. The system will not store a long history of bin status but will operate on real-time data.

Navigation Data: The autonomous vehicle will use real-time navigation data collected through sensors like cameras and LiDAR. This data will be processed locally on the Jetson Nano and used for immediate decision-making, such as navigating to the nearest full bin.

Sensor Data: Information from the ultrasonic sensors on the bins will be transmitted to the Jetson Nano, which will temporarily store this data for current waste collection operations.

The system will not include any external databases at this stage. All data will be stored and processed locally on the Jetson Nano, with a focus on real-time functionality rather than long-term storage. Future expansions may involve integrating a cloud-based solution like AWS for data analysis, but this is beyond the current project scope.

Data Storage Organization: Temporary Files: Temporary data regarding bin status and navigation will be written to local storage on the Jetson Nano, ensuring fast access and processing for autonomous driving and waste collection.

No Persistent Storage: Once the waste collection cycle is completed, the temporary data will be overwritten, ensuring minimal use of storage space. Long-term data retention will not be part of the initial design.

This approach will ensure efficient use of the local storage and processing capabilities of the Jetson Nano while maintaining real-time responsiveness of the autonomous vehicle and the overall waste collection system.

Bibliography

[1] Autonomous mobility-on-demand systems for future urban mobility. *Journal of Future Mobility*, 2015.